

99.co: Scaling a seamless property search experience regionally with Google Cloud

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ABOUT 99.CO

99.co enhances its
search experience to expand
into new markets and
talent acquisition with
Google Cloud.



About 99.co

Founded in 2014, 99.co Group is a property technology business in Singapore and Indonesia that aims to help people discover homes that suit their lifestyle needs and aspirations. Its mission is to improve the property search experience using technology, making it more reliable, easy, and trustworthy for its customers.

Industries: Real Estate & Construction

Google Cloud results

- Reduces
image

Expanding
regional
business

categorization
time from five
minutes to 30
seconds with
Vision AI

- Attracts strong engineering talent pool with sought-after managed services platform
- Quickens business decision-making process with BigQuery
- Reduces gateway report by 50% with Cloud Build

From pricing to location, local ai preference, there's a smorgasbo consider when searching for the 99.co Group (<https://www.99.co/>) (9 the process shouldn't feel like a l chore. With the help of technology, it aims to simplify and make the search experience smoother, faster, and more pleasant.

well-
tech

Location: Singapore

(<https://cloud.google.com/build/>), Cloud



About Onigroup

Business technology specialist Onigroup is a Google Premier Partner that serves thousands of clients across 50 countries. With offices in London, Sydney, Singapore, and Jakarta, its mission is to educate and enable businesses to reduce complexity using advanced tech solutions.

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Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Vision API (<https://cloud.google.com/vision/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Cloud Build (<https://cloud.google.com/build/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Container Registry

(<https://cloud.google.com/container-registry/>)

Operating in Singapore and Indonesia with 20 million monthly active users, it's imperative that 99.co has a reliable platform to cater to its stakeholders. "We want to redesign the user experience in the renting and buying journey through powerful search, layered with other data dimensions that are non-transactional, such as travel time, point of interest, and other attributes that are part of the customer's considerations," says Nhat Quang Nguyen, vice president of engineering at 99.co. To differentiate itself, the company stays away from the conventional "pay-to-list" model, whereby brokers can pay to get more visibility for their properties. Instead, listings on 99.co are classified based on their quality.

The company has been a long-term Google Cloud (<https://cloud.google.com/>) customer and has been working closely with Onigroup (<https://onigroupglobal.com/>) since 2016, expanding its repertoire of products used to support the business over the years. From design and architecture, assessment and recommendation, to integration guidance on using the full technology stack, the Google Cloud Premier Partner has been the key source of advice for the 99.co team.

"Onigroup really understands our business and sometimes even goes the extra mile to give insights on the latest trends in real estate, something that goes beyond their job scope. This shows their high level of commitment to this partnership, which is something we value," adds Nguyen.

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cloud provider to
Cloud Storage
and served them
on Cloud CDN,
we reduced our
loading speed by
up to 50%,
significantly
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user experience.”

—Nhat Quang Nguyen,
Vice President of
Engineering, 99.co

Improving the search experience with a locally hosted cloud provider

When the group acquired [Rumah123](https://www.rumah123.com/en/) (https://www.rumah123.com/en/) in Indonesia, it needed to revamp the infrastructure to accommodate the large number of users it had to host. The original technology stack was hosted from servers in Australia, which affected the performance of the platform. 99.co moved the consumer-facing side of the workload to Google Cloud, to be hosted locally in Indonesia. This helped reduce latency in load time by 50%.

“The search experience relies heavily on users being able to browse high-quality photos at various sizes in rapid succession. When we moved Rumah123’s image assets from a different cloud provider to [Cloud Storage](#) (/storage) and served them on [Cloud CDN](#) (/cdn), we reduced our loading speed by up to 50%, significantly improving the user experience,” says Nguyen. Having a faster load time with high-quality photos has improved customer conversion rate by up to 6x in a highly competitive environment.

Attracting a strong talent pool with managed services

Managed databases have become an attractive factor for engineers in recent years because it takes

away the need for maintenance. This allows them more time to experiment and innovate new features to support business needs, which is what makes their jobs more exciting. To keep up with the fast-growing number of daily users in both Singapore and Indonesia, 99.co is focused on growing its talent pool of skilled engineers, specifically DevOps and infrastructure engineers.

“The tech space is a competitive one, and to attract the top talent in the industry, we need to have the best tools in the industry. [Google Kubernetes Engine \(/kubernetes-engine\)](#), [Cloud SQL \(/sql\)](#), [Cloud Vision \(/vision\)](#), [Cloud Build \(/cloud-build\)](#), and [Container Registry \(/container-registry\)](#) are some of the tools we use that are very much sought after, helping us in our talent acquisition,” explains Nguyen. He adds that Google Cloud products integrate well with popular community solutions, which allows engineers to leverage their existing skill sets.

“Many potential employees from the engineering side have reached out to us, mentioning that they were keen because of the tech stack we use. This is because Google Cloud has the most advanced, highly sought after combination of products, and it has really allowed us to select the best of the best.”

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Scaling up at speed with Cloud SQL and BigQuery

Moving to [BigQuery](#) (/bigquery) has democratized the way employees access data at 99.co, as different engineers can run the data they need in parallel. Previously, there was always a bottleneck with data

analysts because every time the team needed to pull specific data, they would have to approach a data analyst to get them. BigQuery allows anyone with basic knowledge of the platform to access the data needed immediately, resulting in faster business decisions.

With most of its data being non-relational data, categorizing it can be a challenge. However, this has been a breeze on BigQuery. “The query language of BigQuery is so easy to understand, and has the flexibility as well as the functions we need. This enables our data analysts to explore and experiment with various datasets more frequently, so we can extract more value out of our data,” says Nguyen.

With the help of Cloud SQL, developers can also get a better understanding by doing a deep dive into the data without relying on the information provided by data analysts. “On one hand, developers have a lot more freedom and flexibility, while on the other hand, data analysts now have more time to answer questions from the sales team and better equip them with the information they need,” explains Nguyen.

BigQuery also acts as the organization’s go-to managed database service. With the load managed by Cloud SQL, engineers don’t have to worry about writing manual backups or hiring a database specialist just to maintain it. Instead, 99.co can direct more manpower to product development instead.

Building a smarter platform with Vision AI

To remain one step ahead of the industry, 99.co uses Vision AI to automatically categorize the photos taken by agents directly on its platform. This has been a huge time saver for agents as the photos they take can be uploaded and put in the right order automatically. “In the past, agents would have to key in the categories and label each photo that is uploaded with the metadata required,” Nguyen explains. With around 15 to 20 photos per listing, this used to be a laborious and time-consuming process.

Because of the machine learning abilities of Vision AI, each photo is now automatically categorized and labeled. For example, if an image has a television and a couch in it, the system would recognize it as being a living room and automatically label it in the system. It also knows that photos with people in them tend to get lower traction, so if a photo has a human face, it would automatically reject that photo. “What used to take three to five minutes can now be done within 30 seconds, and given that our agents manage anywhere from 10 to 100 listings per day, this feature is a huge time saver,” says Nguyen.

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Speeding up internal services with Cloud Build

The 99.co team recently started experimenting with Cloud Build to speed up its internal processes to accommodate the fast-growing business. "As the number of services grows, we've noticed that our build pipeline is always busy, especially during peak release hours," says Nguyen. "Since our client builds

can take an average of 20 to 30 minutes to complete, we've experienced many bottlenecks, forcing teams to manually prioritize." To solve this problem, 99.co has been experimenting with Cloud Build, which has a serverless instructure. This has allowed a much quicker build, reducing the time to just two minutes. "The early results on Cloud Build have been very promising, with our main backend server build dropping from four minutes to just 20 seconds from Travis to Cloud Build. We can't wait to see how much more time we will save, especially on bigger builds," says Nguyen.

With so much in store, 99.co is excited to continue working with Google Cloud and Onigroup as it continues to expand regionally. As Nguyen says, "We already have plans to expand into another Southeast Asian market in 2022. With the successes we've had with Google Cloud, I believe that we will be able to replicate what we have in the new market and continue experimenting together as we grow."